

Spherical Robot Electromechanics - Harry

Motor Assembly

SR-01

Sub-system control

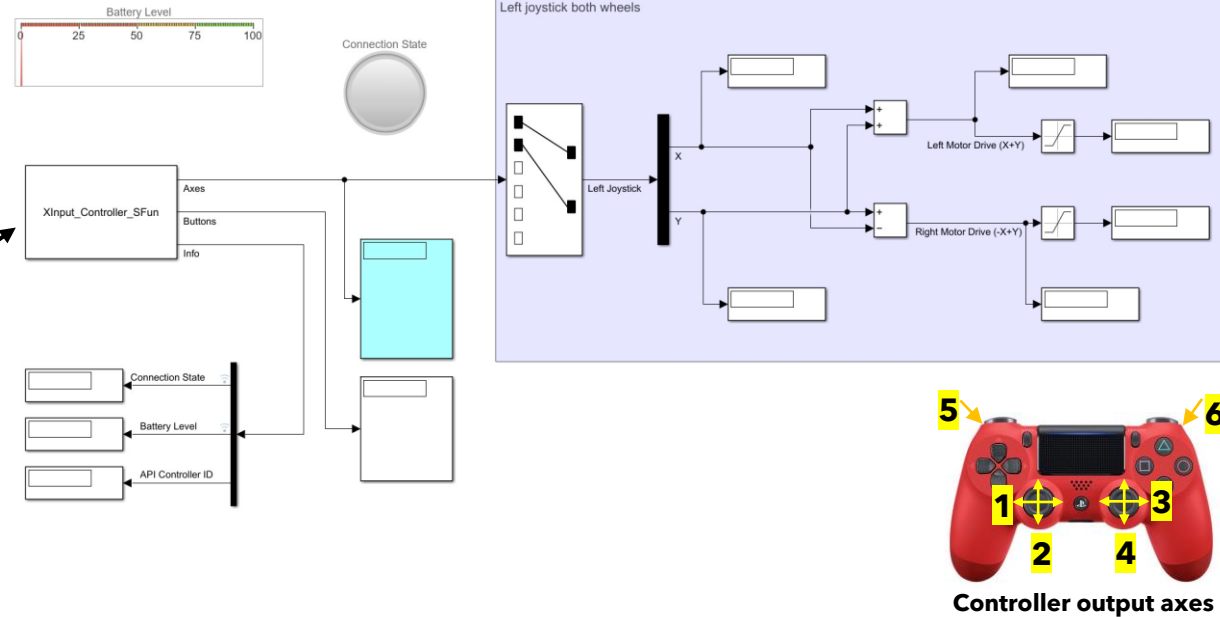
Integrated Sub-system behaves as expected under RC control

Must

Basic RC control direction and speed can be sent to the motor controller (pi) to drive the sphere robot to specific locations

SR-01: **Not tested**

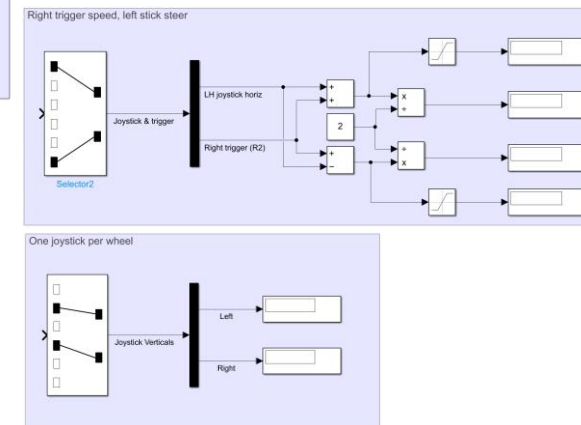
Map game controller inputs to control robot via simulink



To be completed:
23-Oct-2023

Left joystick inputs mapped to left and right motor demanded fraction of max speed. (0→1)

Alternative mappings also considered:



To be done:

- Only step to do is testing game controller inputs to directly control robot (only requires building of complete first prototype - Simulink code already completed)
- Completed by 23-Oct-2023.

SR-02: **Pass**

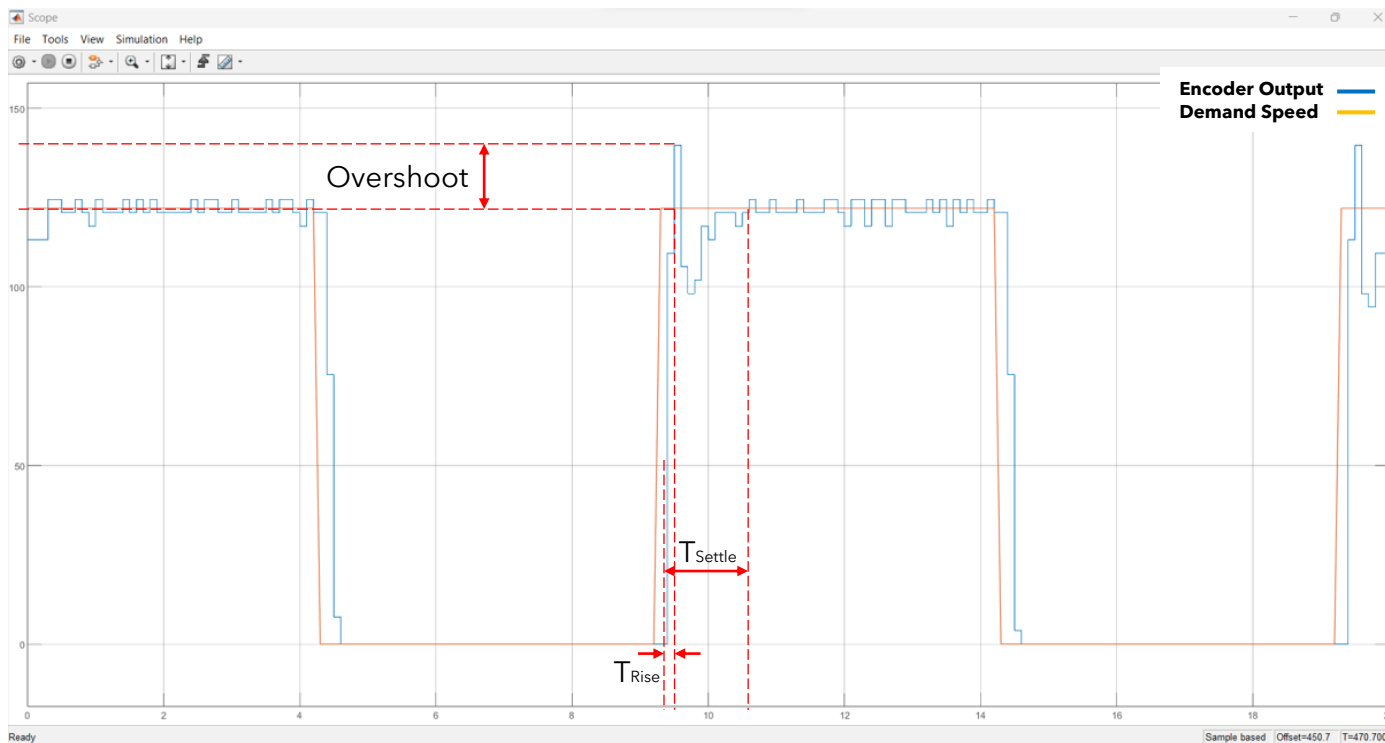
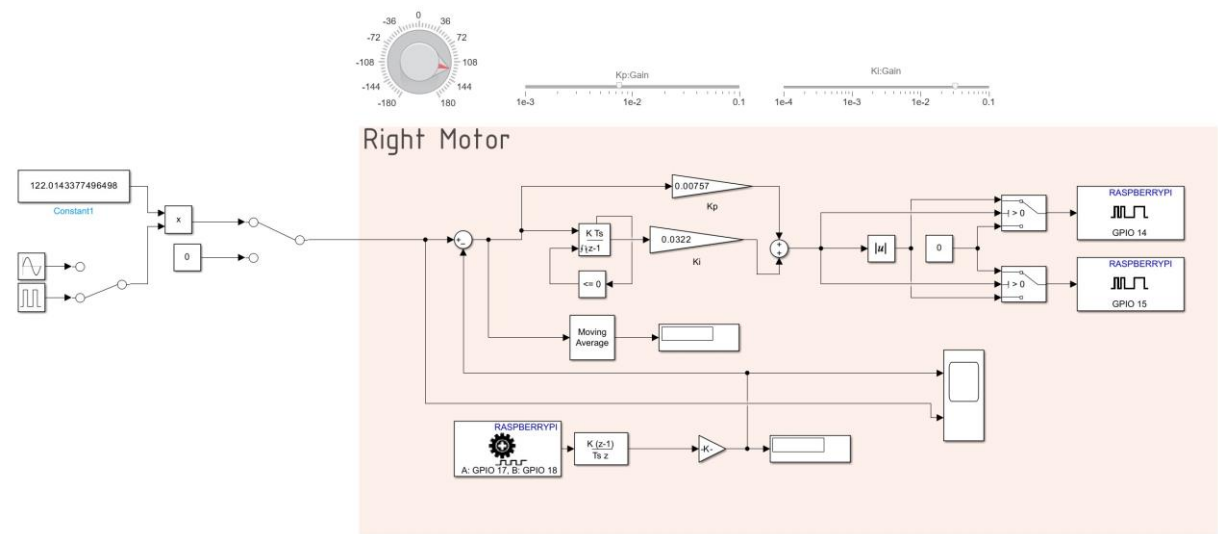
PI control of motor implemented.

Final gain values: Kp: **0.00757**
Ki: **0.0322**

Demand speed: **122 rpm**

- Overshoot: (140 rpm max) **14%**
- Settle Time: **1.25s**
- Rise Time: **0.17s**

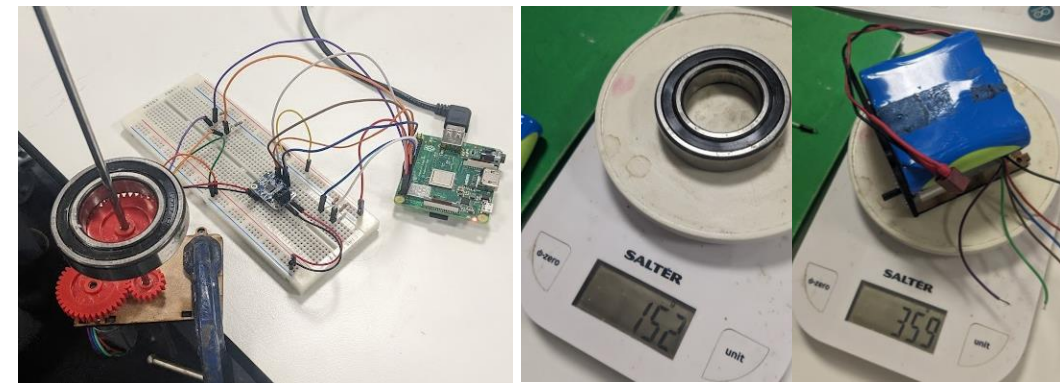
These values were deemed to be acceptable at this stage - further tuning could be done during the integration stage



Test setup - to simulate added friction and an inertial load the motor was mounted to drive a couple of gears and a heavy bearing.
Bearing inertia: $3.64 \times 10^{-4} \text{ kg.m}^2$. Overall gear ratio: **1:25**

$$\text{Reflected inertia: } \frac{3.64 \times 10^{-4}}{25^2} = \mathbf{5.8 \times 10^{-7} \text{ kg.m}^2}$$

This inertia acts as the differential term in the control loop. The bearing was chosen as it is a similar weight to half of the motor chassis (main components: 2x motors and battery, weighed below right). Providing a representative inertia (at least good enough for initial tuning)



SR-03	Motor Drivers	PWM signal from the motor controller drives each motor via the motor driver	Must	PWM signal can be varied to adjust motor output speed and direction (open-loop)
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SR-03: **Pass**

Open-loop control of motor successfully implemented. Motor speed was successfully varied in open-loop control.
Encoder output in Simulink compared to video of motor output shaft to confirm function of the encoder.

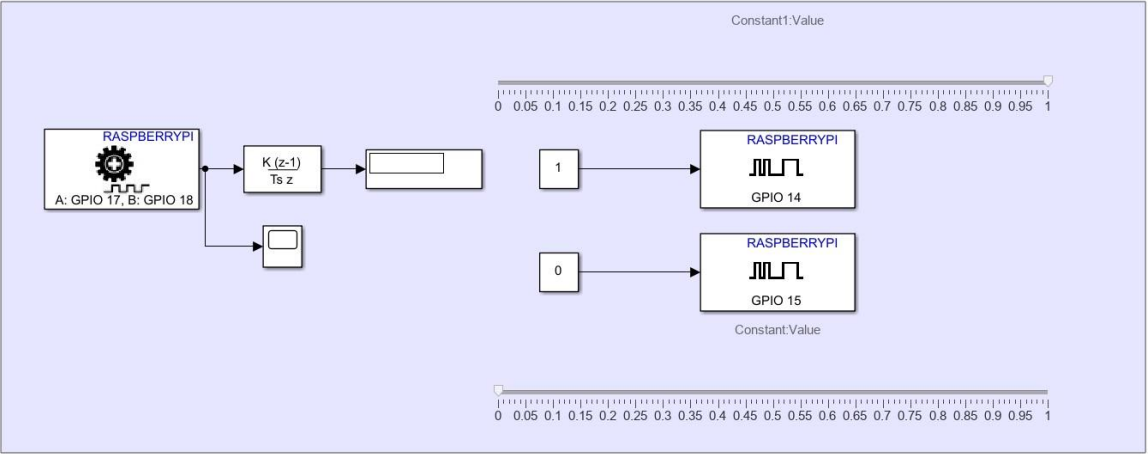
Encoder reading (7.5V power to motor, no PWM): **181rpm**

Average secs/rev = 0.33
 = **3 rps**
 = **180 rpm**

rev. #	revolution time (s)
1	0.78
2	1.12
3	1.45
4	1.78
5	2.10
6	2.43
7	2.78
8	3.10
9	3.43
10	3.75



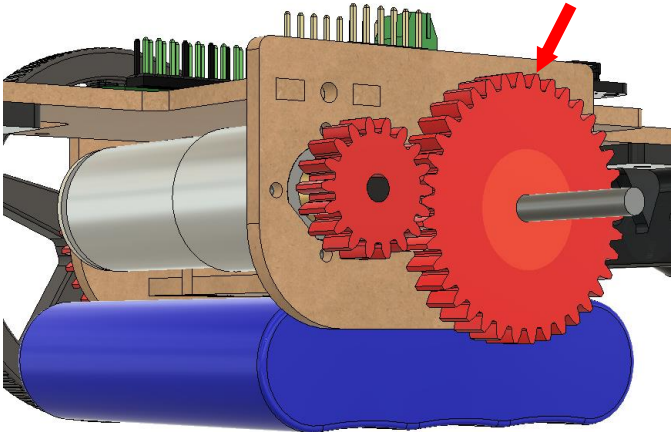
Motor with flag for visual measurements



SR-04	Drive Wheels	Motors are mechanically linked to drive wheels	Must	Motors drive the wheels without putting excessive strain on motor bearings (i.e: flexible/soft coupling, or belt driven)
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SR-04: **Pass**

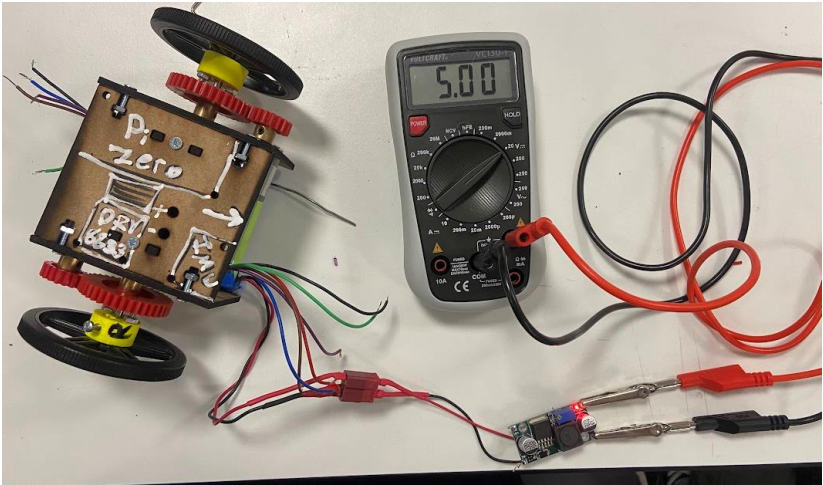
Nylon gears link motor shaft to wheel shaft. These will be the first point of failure since they are a softer material than anything else in the drivetrain



SR-05	Drive Tyres	Drive wheels have sticky 'tyres'	Must	Under normal drive conditions observe that wheels do not spin against the inside of the shell
SR-05: Not tested Drive wheels have rubber tyres (see image). Test method: drive ball a set distance (e.g.: 5 metres), determine theoretical motor rotations (encoder pulses), find error between actual measured encoder pulses vs theoretical no-slip situation. Acceleration can be increased gradually to find limit of grip. Also to be tested: correcting fluid on wheels & shafts to confirm that drive wheels are not slipping on shafts (see picture). Confirm that drive wheels do not slip on drive shaft.				
				To be completed: 23-Oct-2023 
SR-06	Motor Feedback	Motor speed/position is fed back to the Raspberry Pi to enable closed-loop control	Must	Test encoder output. Compare demanded speed to actual speed by averaging over 20 rotations
SR-06: Pass See SR-03 for output speed (encoder) against demand speed test results.				
Power Assembly				
SR-07	Power Converter	Regulated 5V power supplied to Raspberry pi	Must	DC-DC Buck converter regulates 7.4V battery voltage to 5V supply for Pi (check with multimeter)

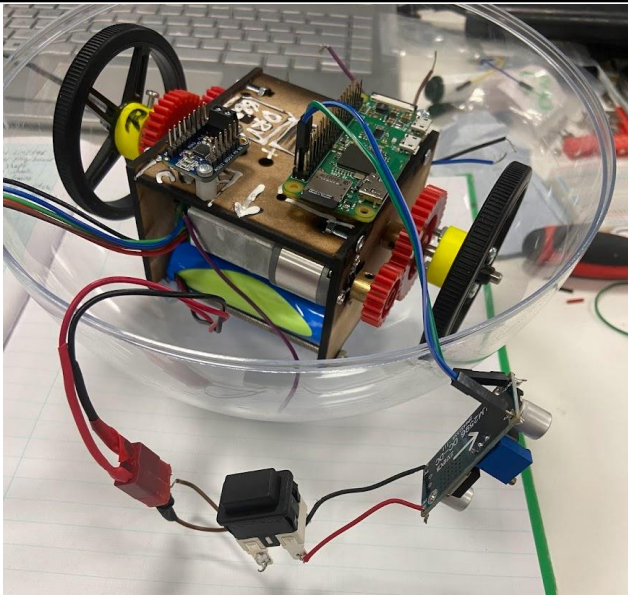
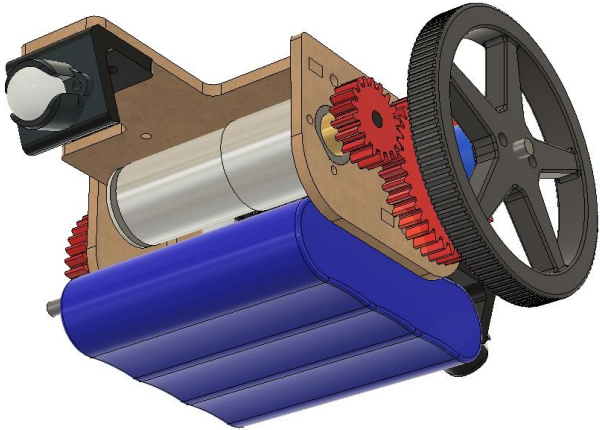
SR-07: **Pass**

An LM2596 Buck Converter module is used to step down the 7.4V nominal battery voltage, to the required 5V to power the Raspberry Pi Zero on-board the sphere robot.



SR-08	Battery Charging	Batteries can be charged without removal	Wish	Design allows access to battery cable without major disassembly of robot
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SR-08: **Pass**
 Battery is externally mounted to robot chassis for easy removal/charging.



SR-09	Battery Maintenance	Batteries can be removed or swapped easily	Wish	Design incorporates easy access to batteries for removal
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SR-09: **Pass**
 See SR-08 above. Battery is held externally on the robot chassis and can be removed without undoing any fasteners or panels.

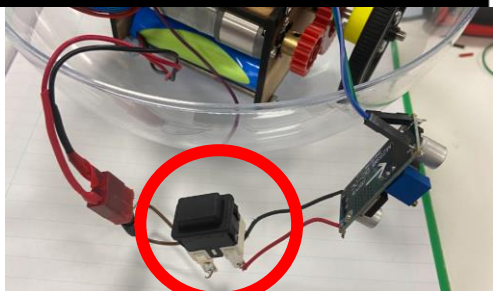
SR-10	Battery Voltage	Feedback given to user of battery voltage state	Wish	Indicator LED or display indicates when battery voltage is low
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SR-10: **Not Implemented**
 Not implemented as of 20-Oct-2023. This will be a small amount of work to implement.

To be completed:
23-Oct-2023

SR-11	On/off switch	Robot includes convenient on/off switch to save power	Wish	Accessible on/off switch included in design, no dismantling required
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SR-11: **Pass**
 Power switch included in initial prototype to allow convenient on/off to Raspberry Pi Zero, and prevent voltage spikes when plugging in/out power to the Raspberry Pi Zero. See image in SR-08 evidence.



SR-12	Low Centre of Gravity	Heavy components are placed low down to increase stability	Must	Robot self-rights when upside down (with power off)
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SR-12: **Pass**

Centre of gravity analysis in CAD shows the CoG is at a radius of 34mm (35%) from the centre of the ball. This allows primary movement of the sphere since moving the offset centre of mass will create a torque (see below). This also promotes stability, and the robot will self-right more readily

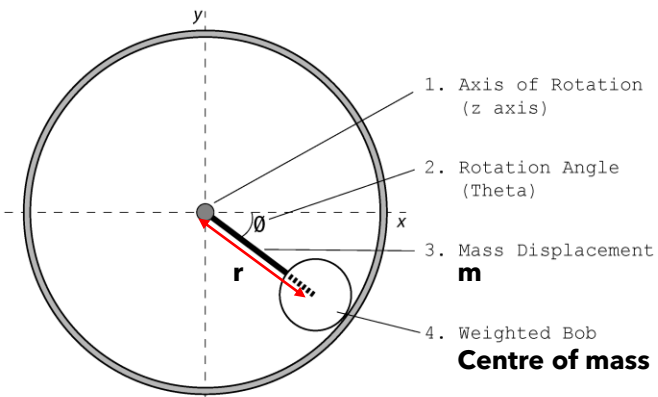
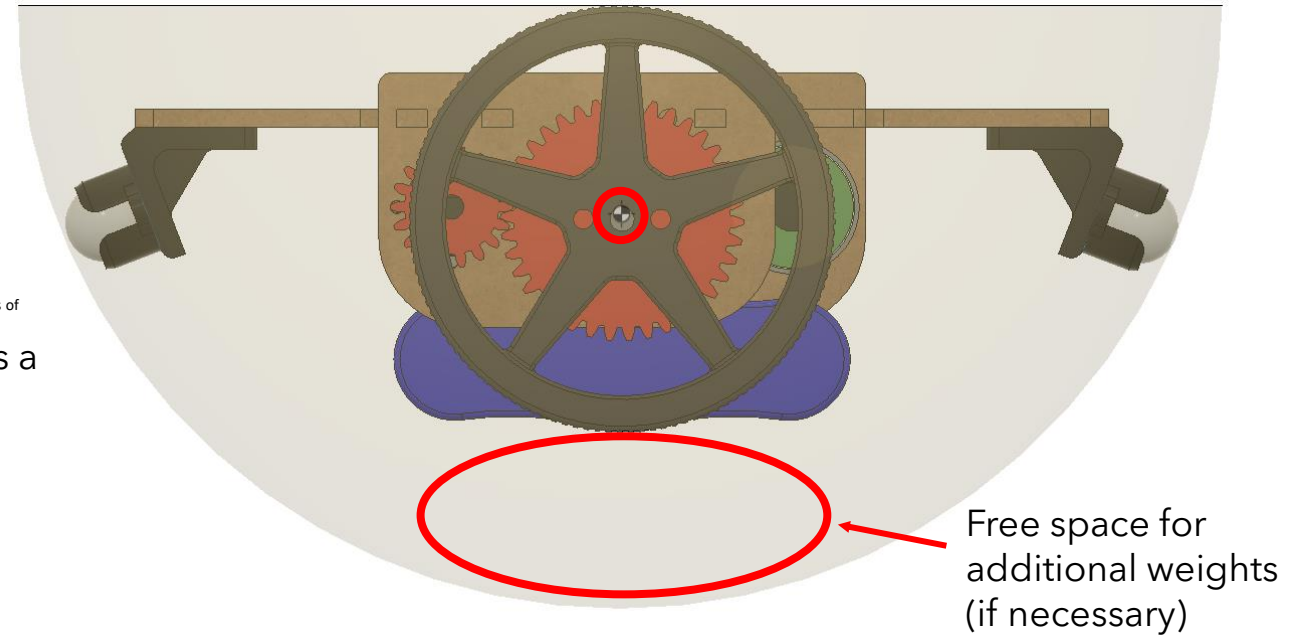


Image from: Chase, R.; Pandya, A. A Review of Active Mechanical Driving Principles of Spherical Robots. *Robotics* **2012**, 1, 3-23. <https://doi.org/10.3390/robotics1010003>

The offset centre of mass creates a torque given by:

$$\tau = mgr \cdot \cos \theta$$

Which drives the robot forward.



The Centre of Gravity being directly through the drive axel may cause problems with the robot not properly righting if it flips upside down inside the ball. Physical testing of the first prototype will confirm this – however, it could be rectified by attaching weights to the bottom of the chassis in the space indicated in the image.

SR-13	ArUco positioning	ArUco is placed flat on top of the mechanism and robot stays upright	Must	Confirm Vision system can always see Robot ArUco when inside the test area
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SR-13: **Not Implemented**

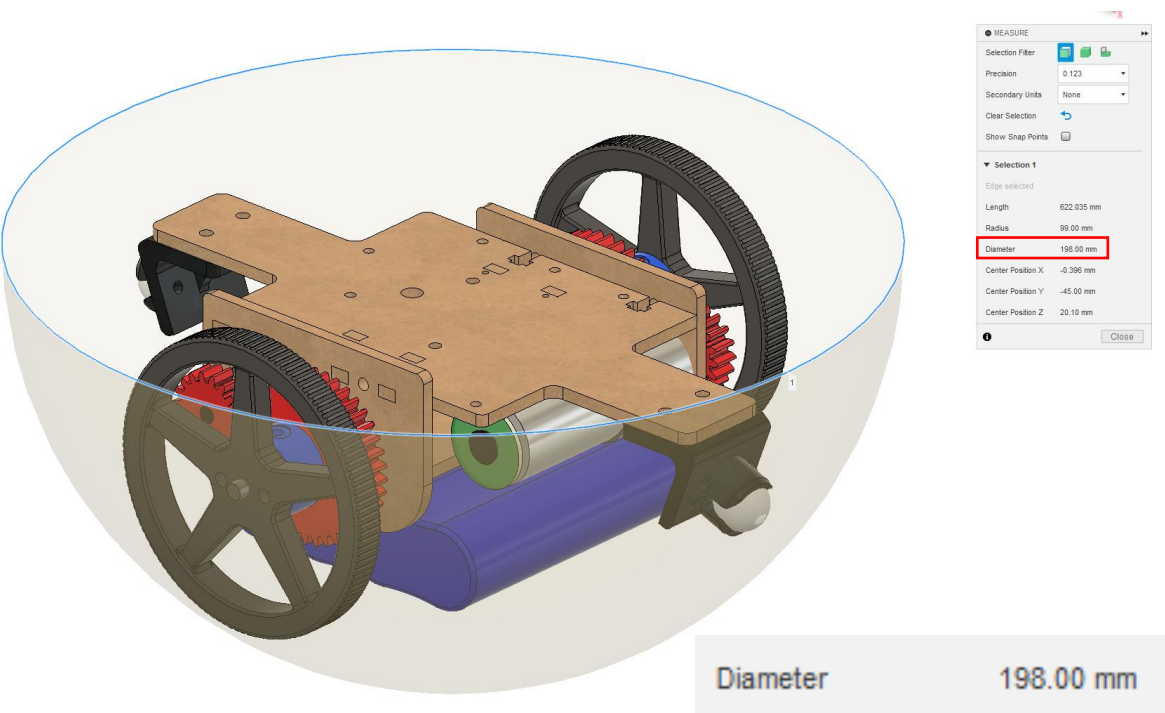
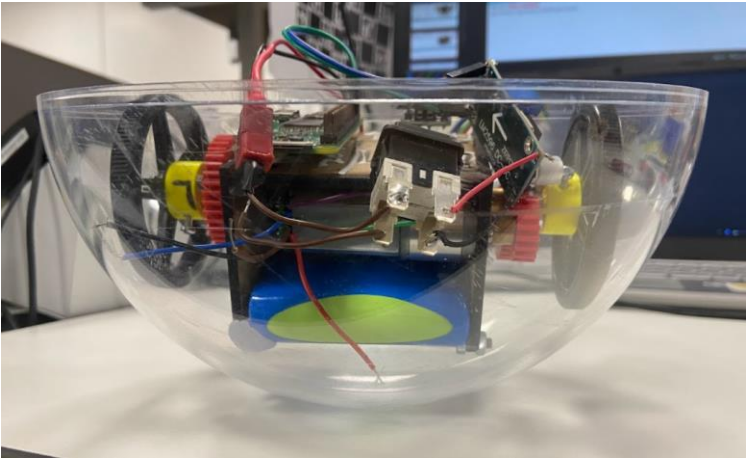
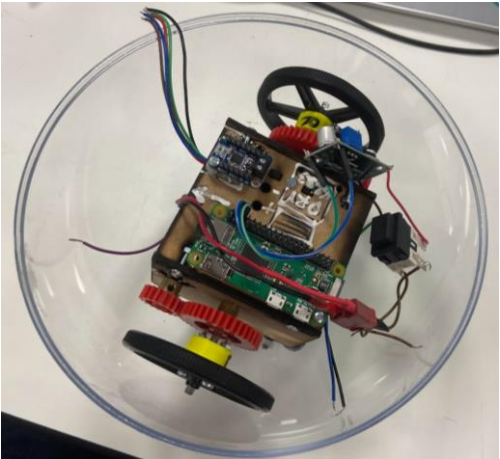
ArUco mounting plate should be pointing upright and not obscured by any of the chassis. IMU feedback will increase stability during driving/stopping – see mission control requirements (Max). To be completed with building of second prototype (midweek w/c 23-Oct-2023)

To be completed:
23-Oct-2023

SR-14	Size	Fits inside a 198mm (internal diameter) plastic ball with tolerance	Must	Analyse CAD model, allow for ± 2 mm of internal diameter variation
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SR-14: Pass

CAD model designed and fitted into a virtual 198mm sphere and confirmed that nothing crossed the surface. (Second iteration CAD shown)
 Physical model (first prototype) placed into sphere shows correct sizing. Casters will be implemented in the second prototype (w/c 23rd October 2023)



SR-15	Friction	Drive wheels/outer shell do not spin against the ground	Must	Observe whether the outer shell spins against the floor. Reduce max demanded acceleration or increase robot weight if necessary.
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SR-15: Not tested

SR-05 merged with SR-15 to do wheel spin/robot grip test. See SR-05 for test/schedule.

To be completed:
23-Oct-2023

Control				
SR-16	Oscillation damping	Robot (and ArUco) stays upright via active control mechanism	Wish	Closed-loop feedback with an IMU is used to actively adjust robot position to keep upright and damp oscillations

SR-16: **Not tested**
Integration with Mission Control (Max) required. Task for w/c 23rd October 2023

Requirement is not my responsibility -
Max is interfacing with the IMU

SR-17	Turning radius	Robot can manoeuvre withing the test area (2m x 2m)	Must	Turning radius <0.5 m
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SR-16: **Not tested**
Testing with game controller as in SR-01 will confirm the robot turning radius. To be completed in line with SR-01 schedule.

To be completed:
23-Oct-2023